Exercise 14

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Introduction to Electrodynamics

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Solution

Verified

Step 1

We need to find the electric field inside a sphere which carry charge density of $\rho = kr$. Consider a spherical Gaussian surface with radius of s and length of l as shown in the following figure. Apply Gauss'e law,

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

the electric field is constant throughout the Gauss's surface, so we can pull the electric field out of the integral, and the integration over the surface equals the surface area is $4\pi r^2$, so,

$$E \cdot 4\pi r^2 = \frac{Q_{\text{enc}}}{\epsilon_0}$$

the enclosed charge equals the integration of the density over the volume of the surface, that is,

$$\begin{aligned}
Q_{\text{enc}} &= \int \rho d\tau \\
&= \int_0^{2\pi} \int_0^{\pi} \int_0^{r} (k\bar{r}) \left(\bar{r}^2 \sin\theta d\bar{r} d\theta d\phi\right) \\
&= 4k\pi \int_0^{r} \bar{r}^3 d\bar{r} = \frac{4\pi k r^4}{4} \\
&= \pi k r^4
\end{aligned}$$

so,

Result

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\pi k r^2 \hat{\mathbf{r}}$$

$$\boxed{\mathbf{E} = \frac{1}{4\pi\epsilon_0}\pi k r^2 \hat{\mathbf{r}}}$$

where the direction of the field is radially outward from the sphere.

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